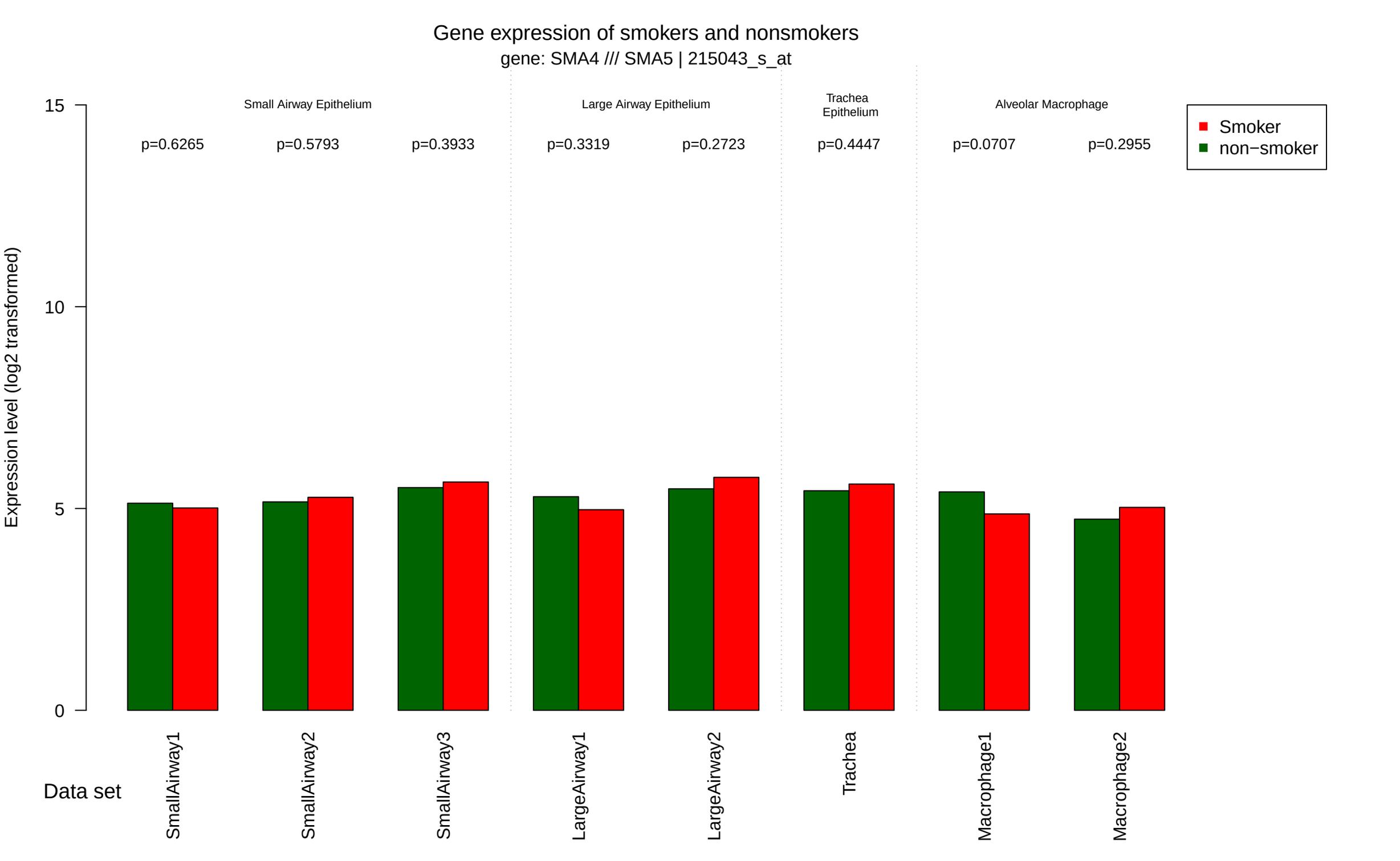
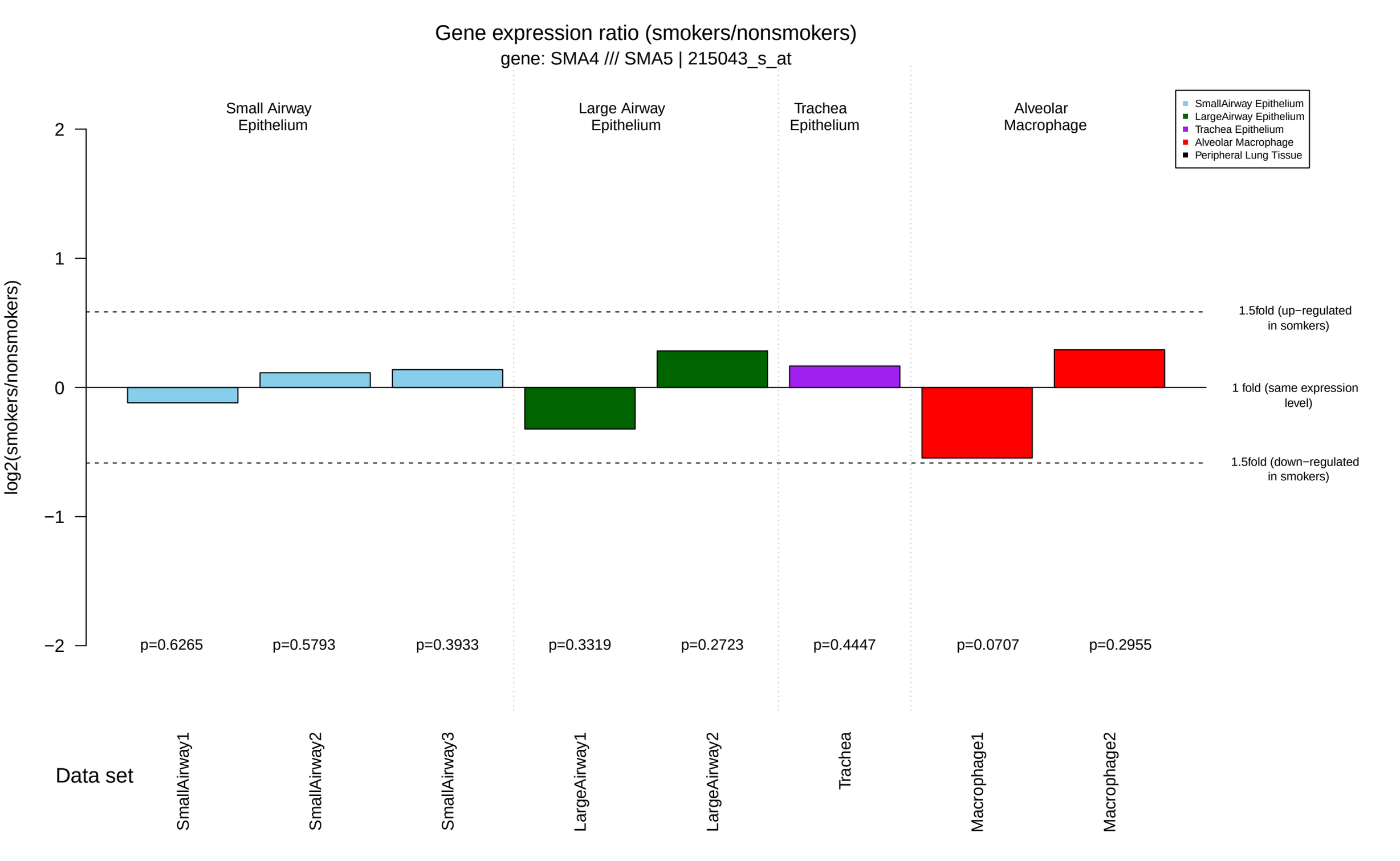
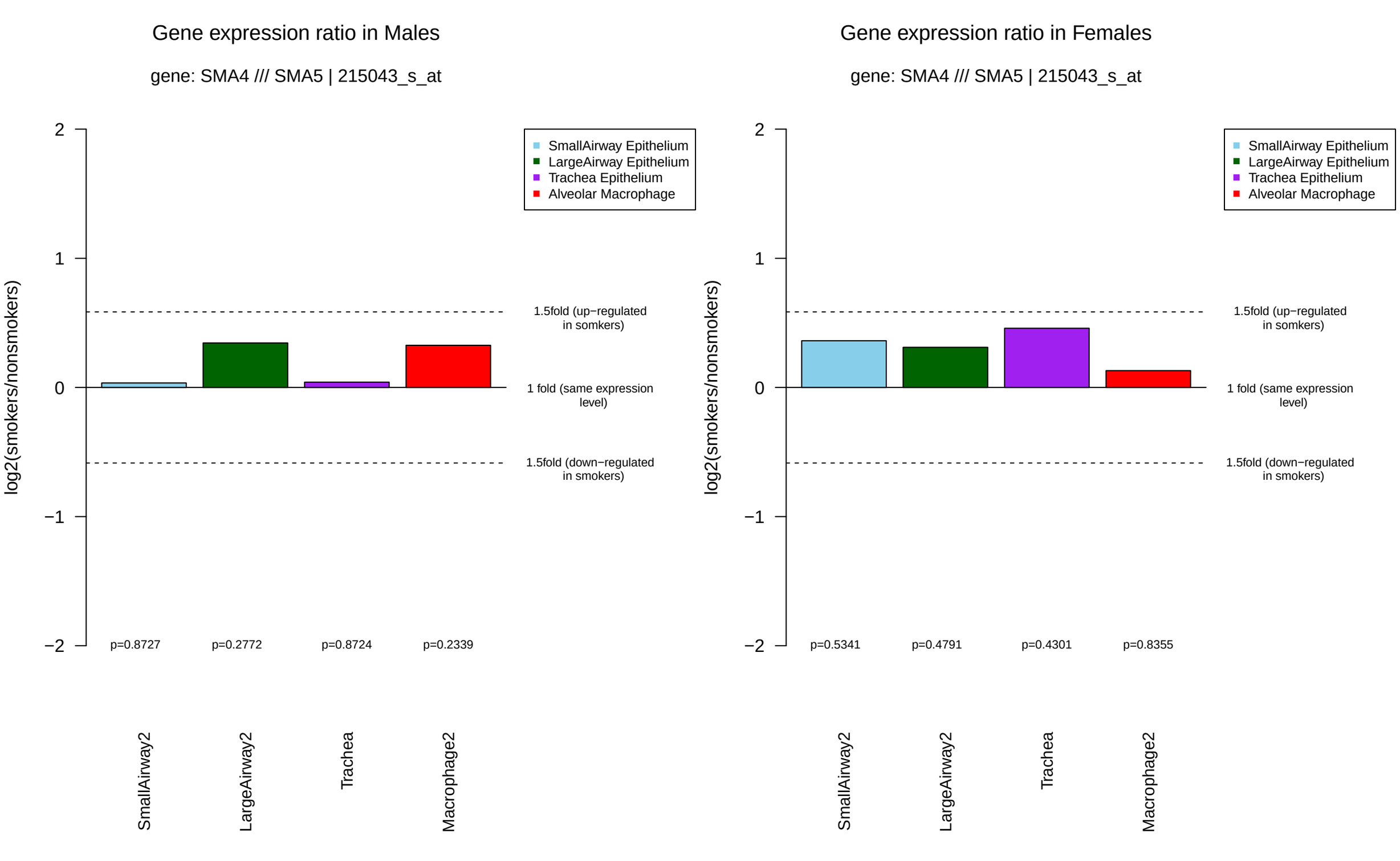
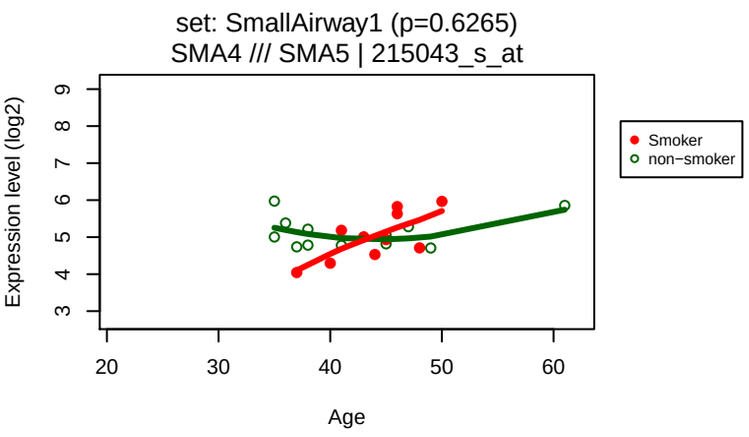
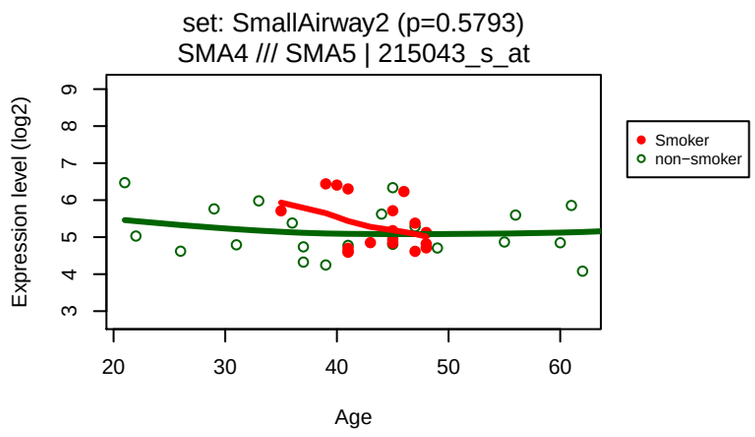
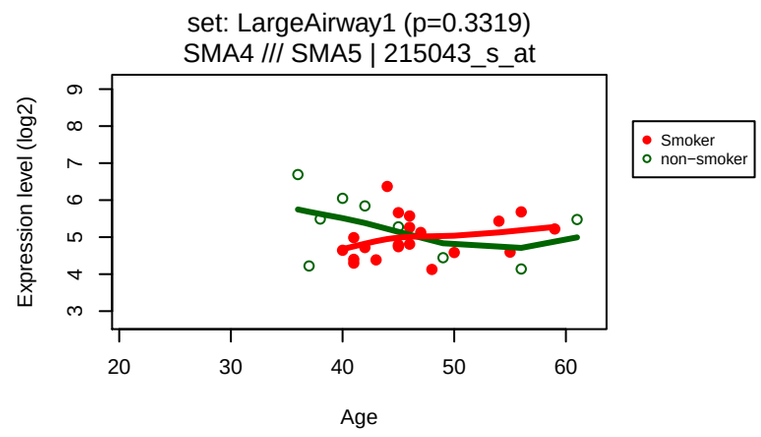

Gene expression of smokers and nonsmokers
gene: SMA4 /// SMA5 | 215043_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6265
p=0.5793
p=0.3933
p=0.3319
p=0.2723
p=0.4447
p=0.0707
p=0.2955
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

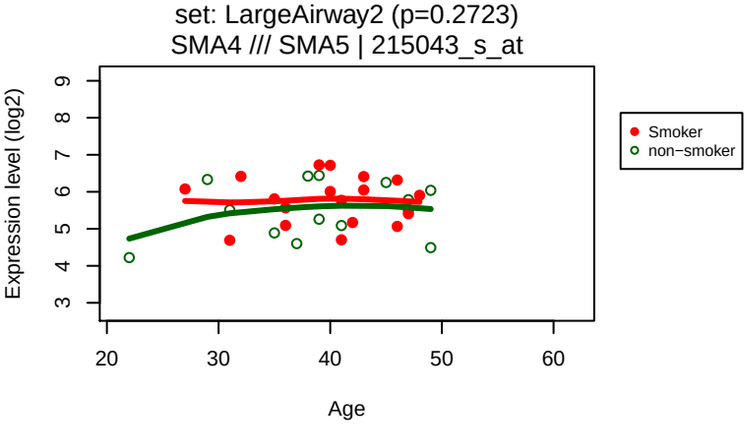

Gene expression ratio (smokers/nonsmokers)
gene: SMA4 /// SMA5 | 215043_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6265
p=0.5793
p=0.3933
p=0.3319
p=0.2723
p=0.4447
p=0.0707
p=0.2955
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

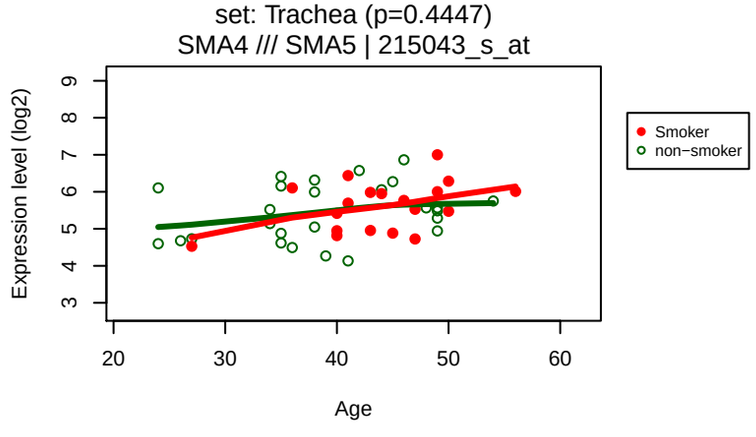

Gene expression ratio in Males
gene: SMA4 /// SMA5 | 215043_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8727
p=0.2772
p=0.8724
p=0.2339
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMA4 /// SMA5 | 215043_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5341
p=0.4791
p=0.4301
p=0.8355
SmallAirway2
LargeAirway2
Trachea
Macrophage2

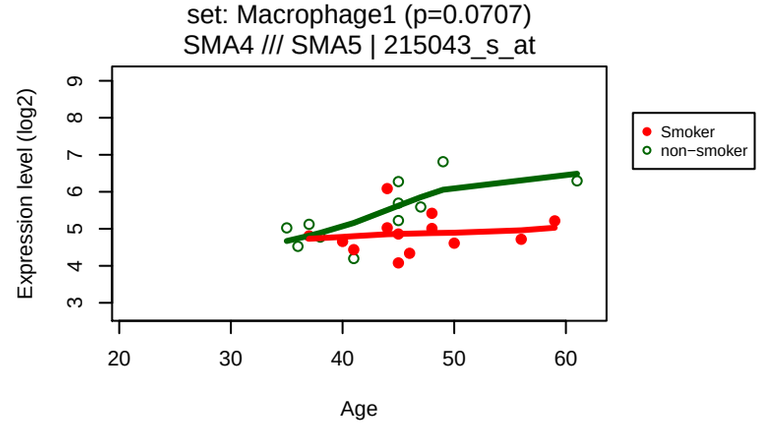

set: SmallAirway1 (p=0.6265)
SMA4 /// SMA5 | 215043_s_at
Expression level (log2)
Age
Smoker
non−smoker

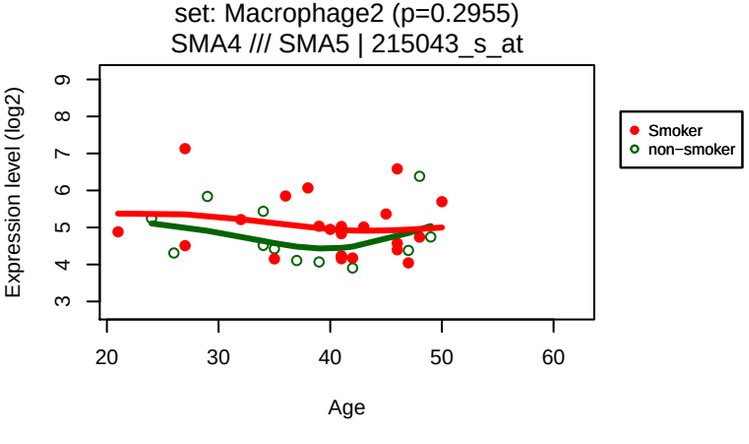

set: SmallAirway2 (p=0.5793)
SMA4 /// SMA5 | 215043_s_at
Expression level (log2)
Age
Smoker
non−smoker

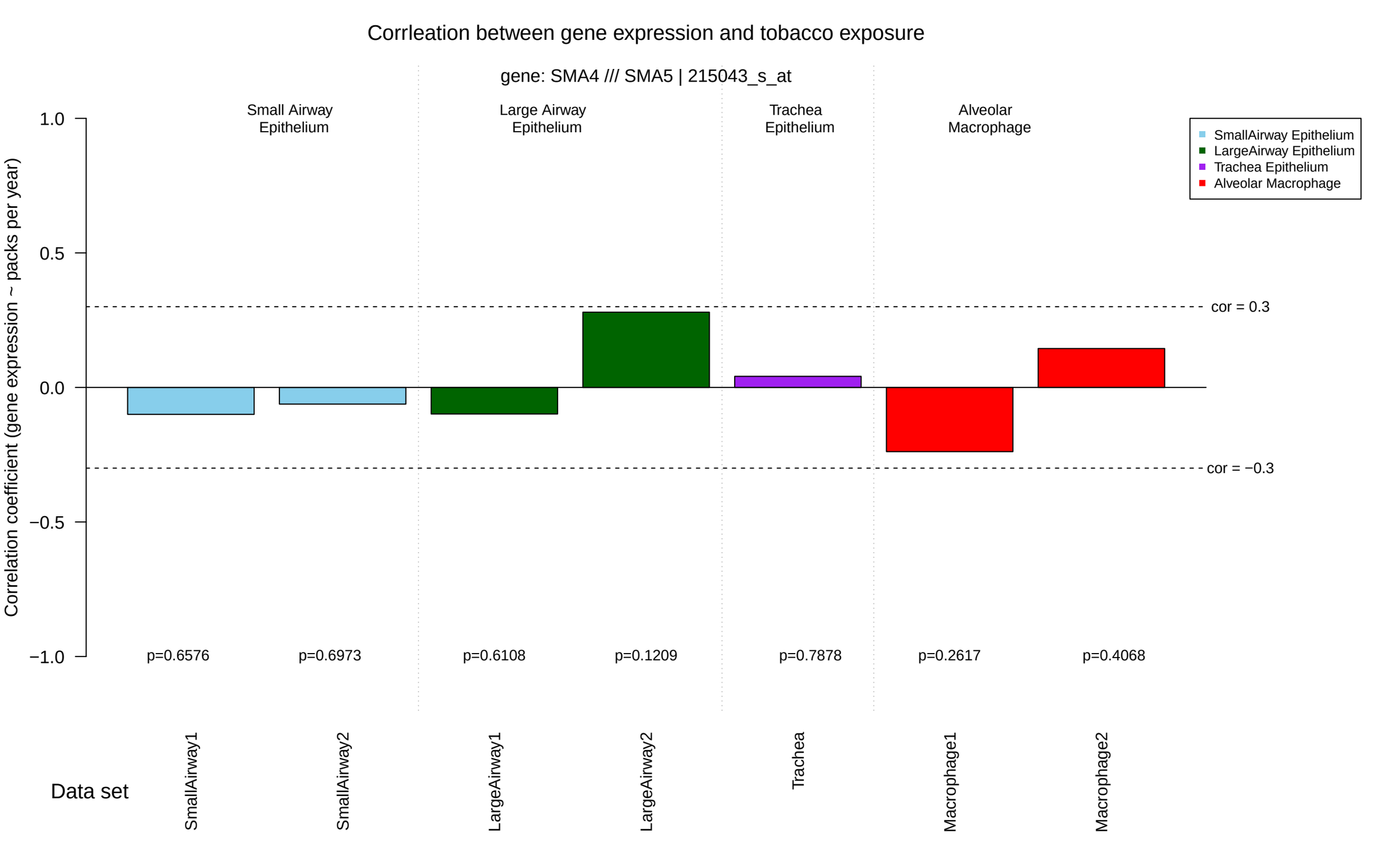

set: LargeAirway1 (p=0.3319)
SMA4 /// SMA5 | 215043_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2723)
SMA4 /// SMA5 | 215043_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4447)
SMA4 /// SMA5 | 215043_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0707)
SMA4 /// SMA5 | 215043_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2955)
SMA4 /// SMA5 | 215043_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMA4 /// SMA5 | 215043_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6576
p=0.6973
p=0.6108
p=0.1209
p=0.7878
p=0.2617
p=0.4068
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2